Petrophysics Lecture 3: Fluid Saturation & Archie Equation

Electrical Resistivity & Hydrocarbon Estimation

Department / Year: Petroleum Geosciences Department - 2nd Year

Lecture Key Outlines & Core Competencies:

- 1. Formation Factor and Tortuosity
- 2. Resistivity Index and Saturation Exponent (Archie Equation)
- 3. Water Saturation (Sw) Determination
- 4. Archie Shaly-Sand Extensions (Waxman-Smits & Dual-Water)

Academic Notice & Verification:

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Support: abdelrahmanelbortogaly@gmail.com | Integrated with Chrome PDF Viewer

Section 2: Practical Calculations, Laboratory Equations & Exercises

Core Equations & Petrophysical / Geological Relations:

Equation 1: Phi_total = (V_bulk - V_grain) / V_bulk * 100%

Equation 2: Q = (k * A * delta_P) / (mu * L) [Darcy Law 1D]

Equation 3: Sw = ((a / Phi^m) * (Rw / Rt))^(1 / n) [Archie Saturation]

Equation 4: V_shale = (GR_log - GR_clean) / (GR_shale - GR_clean) [Linear Index]

Laboratory Work Checklist & Submission:

[OK] Step 1: Calibrate electronic calipers and weigh dry clean core plugs.

[OK] Step 2: Record helium injection pressure and stabilization chamber readings.

[OK] Step 3: Plot data on semi-log scale graph and determine trend line slopes.

[OK] Step 4: Upload your lab report directly via PETROGEO Admin & Student portal.